

Partial Fractions Notes

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1 Introduction

What is the idea behind partial fractions? What if we are tasked with integrating a function like

$$\int \frac{1}{x^2 + 3x + 2} dx?$$

Just looking at this integral, it is not immediately clear how to proceed. However, what if we were instead tasked with integrating something of the form

$$\int \frac{1}{x+1} - \frac{1}{x+2} dx?$$

This is much easier to integrate: just split the integral into two parts and integrate each one separately. This is the idea behind partial fractions: we can take a complicated rational function (a polynomial divided by a polynomial) and break it down into simpler fractions that are easier to integrate directly.

How do we do this? For this derivation, we will leave the integral behind and focus on the algebraic manipulation. We will start with our original function

$$\frac{1}{x^2 + 3x + 2}.$$

The first step is to factor the denominator. This can be done using stuff like the rational root theorem, or if you're clever enough, just guessing. In this case, we can see that the denominator factors as

$$\frac{1}{(x+1)(x+2)}.$$

Where do we go from here? We want to break this down into simpler fractions, but we don't know what those fractions are yet. We can make an educated guess, though. We know that the denominator is a product of two linear factors, so we can guess that the numerator will be a sum of two fractions with linear numerators:

$$\frac{A}{x+1} + \frac{B}{x+2}.$$

From here, we assert that these two expressions are equal, giving us

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}.$$

Now, we can multiply both sides by the denominator to get rid of the fractions:

$$1 = A(x+2) + B(x+1).$$

From here, there are two ways to proceed.

1. The first of which is using a clever trick, my personal favorite. Right now we have the equation

$$1 = A(x + 2) + B(x + 1).$$

However, there are two variables in this equation, A and B . But what if we could eliminate one of the variables? Then we would be left with a single variable equation, which is much easier to solve. We can do this by plugging in a value for x that makes one of the terms zero. For example, if we plug in $x = -2$, then the A term will vanish, leaving us with

$$A(-2 + 2) + B(-2 + 1) = 1$$

$$0 + B(-1) = 1$$

$$B = -1.$$

We can do the same thing with $x = -1$ to get

$$A(-1 + 2) + B(-1 + 1) = 1$$

$$A(1) + 0 = 1$$

$$A = 1.$$

Now we have $A = 1$ and $B = -1$, so we can write our original function as

$$\frac{1}{x^2 + 3x + 2} = \frac{1}{x + 1} - \frac{1}{x + 2}.$$

The unfortunate part of this method is that it requires some cleverness and a bit of luck, as not all equations can be solved this way. However, it is a very powerful technique that can save a lot of time. If you try to use this method and it doesn't work, you can always fall back on the second method.

2. The second method is a more tedious, but more general approach. We can start with the same equation

$$1 = A(x + 2) + B(x + 1).$$

However, instead of plugging in values for x , we can expand the right-hand side to get

$$1 = Ax + 2A + Bx + B.$$

Now, we can combine like terms to get

$$1 = (A + B)x + (2A + B).$$

If we now compare the coefficients of the left-hand side and the right-hand side, we can set up a system of equations:

$$\begin{cases} A + B = 0 \\ 2A + B = 1 \end{cases}$$

This can be solved in a number of ways, but my preferred method is elimination. We can subtract the first equation from the second to get

$$\begin{cases} A + B = 0 \\ 2A + B - (A + B) = 1 - 0 \end{cases}$$

$$\begin{cases} A + B = 0 \\ A = 1 \end{cases}$$

Now we can plug $A = 1$ back into the first equation to get

$$\begin{cases} 1 + B = 0 \\ A = 1 \end{cases}$$

$$\begin{cases} B = -1 \\ A = 1 \end{cases}$$

We can see that both methods give us the same result, so we can use either one depending on the situation. The partial fractions decomposition of our original function is therefore

$$\frac{1}{x^2 + 3x + 2} = \frac{1}{x + 1} - \frac{1}{x + 2}.$$

2 General Case

What if we have a more complicated function, like

$$\frac{1}{x^3 + 3x^2 + 2x}?$$

We can start by factoring the denominator to get

$$\frac{1}{x(x+1)(x+2)}.$$

Now, we can make our educated guess for the partial fractions decomposition:

$$\frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}.$$

In general, we can see that for each factor, we will need an additional fraction.

What if we have an irreducible quadratic factor, like in the example

$$\frac{1}{x^3 + x}?$$

We can start by factoring the denominator to get

$$\frac{1}{x(x^2 + 1)}.$$

Now, we can make our educated guess for the partial fractions decomposition, however we need to account for the fact that the numerator of the quadratic factor can also have x terms now:

$$\frac{A}{x} + \frac{Bx + C}{x^2 + 1}.$$

A special case of this is when we have a repeated factor, like in the example

$$\frac{1}{x^3 + x^2}.$$

We can start by factoring the denominator to get

$$\frac{1}{x^2(x+1)}.$$

Now, we can make our educated guess for the partial fractions decomposition, making sure to account for the irreducible quadratic factor:

$$\frac{A}{x+1} + \frac{Bx + c}{x^2}.$$

We can do some algebra to get

$$\frac{1}{x^2(x+1)} = \frac{A}{x+1} + \frac{Bx}{x^2} + \frac{C}{x^2} = \frac{A}{x+1} + \frac{B}{x} + \frac{C}{x^2}.$$

This tells that if we have a repeated factor, we need to add an additional fraction for each power of the factor. I will summarize the results on the next page.

3 Summary

1. If we have a linear factor, we need a fraction with a constant numerator. For example, if my denominator has a factor of $(x + 1)$, I need to add a fraction of the form $\frac{A}{x+1}$ to my partial fractions decomposition.
2. If we have an irreducible quadratic factor, we need a fraction with a linear numerator. For example, if my denominator has a factor of $(x^2 + 1)$, I need to add a fraction of the form $\frac{Ax+B}{x^2+1}$ to my partial fractions decomposition.
3. If we have a repeated linear factor, we need a fraction for each power of the factor. For example, if my denominator has a factor of x^3 , I to add fractions of the form $\frac{A}{x}$, $\frac{B}{x^2}$, and $\frac{C}{x^3}$ to my partial fractions decomposition.

4 Examples

4.1 Example 1

Find the partial fractions decomposition of

$$\frac{4x + 6}{x^3 + 4x^2 + 3x}.$$

Solution. First, we can factor the denominator to get

$$\frac{4x + 6}{x(x^2 + 4x + 3)} = \frac{4x + 6}{x(x + 1)(x + 3)}.$$

Now, we can make our educated guess for the partial fractions decomposition:

$$\frac{A}{x} + \frac{B}{x + 1} + \frac{C}{x + 3}.$$

We can multiply both sides by the denominator to get

$$4x + 6 = A(x + 1)(x + 3) + B(x)(x + 3) + C(x)(x + 1).$$

Now, we can plug in values for x to eliminate variables. If we plug in $x = 0$, we get

$$4(0) + 6 = A(0 + 1)(0 + 3) + B(0)(0 + 3) + C(0)(0 + 1)$$

$$3A = 6$$

$$A = 2.$$

If we plug in $x = -1$, we get

$$4(-1) + 6 = A(-1 + 1)(-1 + 3) + B(-1)(-1 + 3) + C(-1)(-1 + 1)$$

$$-2B = 2$$

$$B = -1.$$

If we plug in $x = -3$, we get

$$4(-3) + 6 = A(-3 + 1)(-3 + 3) + B(-3)(-3 + 3) + C(-3)(-3 + 1)$$

$$6C = -6$$

$$C = -1.$$

Now we can plug these values back into our partial fractions decomposition to get

$$\frac{4x + 6}{x^3 + 4x^2 + 3x} = \frac{2}{x} - \frac{1}{x + 1} - \frac{1}{x + 3}.$$

4.2 Example 2

Find the partial fractions decomposition of

$$\frac{x + 2}{x^2 + 2x + 1}.$$

Solution. First, we can factor the denominator to get

$$\frac{x + 2}{(x + 1)^2}.$$

Now, we can make our educated guess for the partial fractions decomposition:

$$\frac{A}{x + 1} + \frac{B}{(x + 1)^2}.$$

We can multiply both sides by the denominator to get

$$x + 2 = A(x + 1) + B.$$

Now, we can plug in values for x to eliminate variables. If we plug in $x = -1$, we get

$$-1 + 2 = A(-1 + 1) + B$$

$$B = 1.$$

If we plug in $x = 0$, we get

$$0 + 2 = A(0 + 1) + B$$

$$A + B = 2$$

We already know that $B = 1$, so we can plug that in to get

$$A + 1 = 2$$

$$A = 1.$$

Now we can plug these values back into our partial fractions decomposition to get

$$\frac{x + 2}{x^2 + 2x + 1} = \frac{1}{x + 1} + \frac{1}{(x + 1)^2}.$$

4.3 Example 3

Find the partial fractions decomposition of

$$\frac{18}{x^2(x-3)}$$

and use the results to integrate

$$\int \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} dx.$$

Solution. As the denominator is already factored, we can begin by making our guess for the partial fractions decomposition:

$$\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-3}.$$

We can multiply both sides by the denominator to get

$$Ax(x-3) + B(x-3) + Cx^2 = 18.$$

Now, we can plug in values for x to eliminate variables. If we plug in $x = 0$, we get

$$0 + B(-3) + 0 = 18$$

$$-3B = 18$$

$$B = -6.$$

If we plug in $x = 3$, we get

$$A(3)(0) + B(0) + C(3^2) = 18$$

$$9C = 18$$

$$C = 2.$$

We can plug these values back into our partial fractions decomposition to get

$$Ax(x-3) - 6(x-3) + 2x^2 = 18.$$

If we expand this out and combine like terms, we get

$$Ax^2 - 3Ax - 6x + 18 + 2x^2 = 18.$$

$$(A+2)x^2 + (-3A-6)x + 18 = 18.$$

This tells us that

$$\begin{cases} A+2=0 \\ -3A-6=0 \end{cases}$$

We can solve this system of equations to get

$$A = -2.$$

Therefore, our partial fractions decomposition is

$$\frac{18}{x^2(x-3)} = \frac{-2}{x} - \frac{6}{x^2} + \frac{2}{x-3}.$$

Now, we look back at the integral

$$\int \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} dx.$$

As the degree of the numerator is greater than the degree of the denominator, we can use polynomial long division to get

$$\frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} = x - 2 - \frac{18}{x^2(x-3)}.$$

Now, we can use our partial fractions decomposition to get

$$\begin{aligned} \int \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} dx &= \int \left(x - 2 - \frac{18}{x^2(x-3)} \right) dx \\ &= \int (x - 2) dx + \int \left(\frac{2}{x} + \frac{6}{x^2} - \frac{2}{x-3} \right) dx \\ &= \frac{x^2}{2} - 2x + 2 \ln |x| - \frac{6}{x} - 2 \ln |x-3| + C. \end{aligned}$$